Database System Engineering and Distributed Backend Development

Course Code: 25CS13002E

ROLL NO:2520030078

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Query 1: Create the database

```
CREATE DATABASE Bank_ACID_DB1;
USE Bank_ACID_DB1;
```

Output: -

#	Time	Action	Message	Duration / Fetch
1	19:06:12	CREATE DATABASE Bank_ACID_DB1	1 row(s) affected	0.031 sec
2	19:06:14	USE Bank_ACID_DB1	0 row(s) affected	0.000 sec

Query 2: -CREATE CUSTOMER TABLE

```
CREATE TABLE Customer (
    Customer_ID INT PRIMARY KEY,
    Customer_Name VARCHAR(100) NOT NULL,
    Phone VARCHAR(15),
    City VARCHAR(50)
);
```

Output: -

3	19:07:15	CREATE TABLE Customer (Customer_ID INT PR...	0 row(s) affected	0.079 sec

Query 3: -CREATE ACCOUNT TABLE

```
CREATE TABLE Account (
    Account_No INT PRIMARY KEY,
    Customer_ID INT,
    Account_Type VARCHAR(20),
    Balance DECIMAL(12,2),
    Branch VARCHAR(50),

    FOREIGN KEY (Customer_ID)
    REFERENCES Customer(Customer_ID)
);
```

Output: -

```
4  19:08:34  CREATE TABLE Account (  Account_No INT PRI...  0 row(s) affected  0.094 sec
```

Query 4: -CREATE TRANSACTION TABLE

```
CREATE TABLE Bank_Transaction (
    Transaction_ID INT PRIMARY KEY AUTO_INCREMENT,
    Account_No INT,
    Transaction_Type VARCHAR(20),
    Amount DECIMAL(12,2),
    Transaction_Date DATETIME DEFAULT CURRENT_TIMESTAMP,

    FOREIGN KEY (Account_No)
    REFERENCES Account(Account_No)
);
```

Output: -

```
5  19:09:46  CREATE TABLE Bank_Transaction (  Transaction...  0 row(s) affected
```

Query 5: -INSERT CUSTOMER DATA

```
INSERT INTO Customer
(Customer_ID, Customer_Name, Phone, City)
VALUES
(101, 'Aswitha Patel', '9876543210', 'Hyderabad'),

(102, 'Adhritha Patel', '9876543211', 'Vijayawada'),

(103, 'Anirudh Reddy', '9876543212', 'Bangalore'),

(104, 'Akshara Reddy', '9876543213', 'Chennai'),

(105, 'Ananya Varshita Reddy', '9876543214', 'Hyderabad');
```

Output: -

```
6  19:12:51  INSERT INTO Customer (Customer_ID, Customer_N...  5 row(s) affected Records: 5  Duplicates: 0  Warning...  0.031 sec
```

Query 6: -INSERT ACCOUNT DATA

```
INSERT INTO Account
(Account_No, Customer_ID, Account_Type, Balance, Branch)
VALUES
(10001, 101, 'Savings', 50000, 'Hyderabad'),

(10002, 102, 'Savings', 75000, 'Vijayawada'),

(10003, 103, 'Current', 120000, 'Bangalore'),

(10004, 104, 'Savings', 45000, 'Chennai'),

(10005, 105, 'Current', 90000, 'Hyderabad');
```

Output: -

✔	7	19:14:08	INSERT INTO Account (Account_No, Customer_I...	5 row(s) affected Records: 5 Duplicates: 0 Warnin...	0.032 sec

Query 7: -INSERT BANK TRANSACTION DATA

```
INSERT INTO Bank_Transaction
(Account_No, Transaction_Type, Amount)
VALUES
(10001, 'DEPOSIT', 10000),

(10002, 'DEPOSIT', 15000),

(10003, 'WITHDRAW', 20000),

(10004, 'DEPOSIT', 5000),

(10005, 'WITHDRAW', 10000);
```

Output: -

✔	8	19:15:22	INSERT INTO Bank_Transaction (Account_No, Tr...	5 row(s) affected Records: 5 D

Query 8: -DISPLAY DATA

```
SELECT * FROM Customer;
```

Output: -

Customer_ID	Customer_Name	Phone	City
101	Aswitha Patel	9876543210	Hyderabad
102	Adhritha Patel	9876543211	Vijayawada
103	Anirudh Reddy	9876543212	Bangalore
104	Akshara Reddy	9876543213	Chennai
105	Ananya Varshita Reddy	9876543214	Hyderabad
NULL	NULL	NULL	NULL

Customer 1 ×

Apply

Output

Action Output

#	Time	Action	Message
4	19:08:34	CREATE TABLE Account (Account_No INT P...	0 row(s) affected
5	19:09:46	CREATE TABLE Bank_Transaction (Transacti...	0 row(s) affected
6	19:12:51	INSERT INTO Customer (Customer_ID, Customer...	5 row(s) affected Re
7	19:14:08	INSERT INTO Account (Account_No, Customer_...	5 row(s) affected Re
8	19:15:22	INSERT INTO Bank_Transaction (Account_No, ...	5 row(s) affected Re
9	19:16:44	SELECT * FROM Customer LIMIT 0, 1000	5 row(s) returned

```
SELECT * FROM Account;
```

Account_No	Customer_ID	Account_Type	Balance	Branch
10001	101	Savings	50000.00	Hyderabad
10002	102	Savings	75000.00	Vijayawada
10003	103	Current	120000.00	Bangalore
10004	104	Savings	45000.00	Chennai
10005	105	Current	90000.00	Hyderabad
NULL	NULL	NULL	NULL	NULL

Account 2 ×

Apply

Output

Action Output

#	Time	Action	Message
5	19:09:46	CREATE TABLE Bank_Transaction (Transacti...	0 row(s) affected
6	19:12:51	INSERT INTO Customer (Customer_ID, Customer...	5 row(s) affected R
7	19:14:08	INSERT INTO Account (Account_No, Customer_...	5 row(s) affected R
8	19:15:22	INSERT INTO Bank_Transaction (Account_No, ...	5 row(s) affected R
9	19:16:44	SELECT * FROM Customer LIMIT 0, 1000	5 row(s) returned
10	19:17:00	SELECT * FROM Account LIMIT 0, 1000	5 row(s) returned

```
SELECT * FROM Bank_Transaction;
```

Transaction_ID	Account_No	Transaction_Type	Amount	Transaction_Date
1	10001	DEPOSIT	10000.00	2026-08-15 19:15:22
2	10002	DEPOSIT	15000.00	2026-08-15 19:15:22
3	10003	WITHDRAW	20000.00	2026-08-15 19:15:22
4	10004	DEPOSIT	5000.00	2026-08-15 19:15:22
5	10005	WITHDRAW	10000.00	2026-08-15 19:15:22
NULL	NULL	NULL	NULL	NULL

Bank_Transaction 3

Output

Action Output

#	Time	Action
6	19:12:51	INSERT INTO Customer (Customer_ID, Customer_Name, Phone, City) VALUES (101, 'Asw
7	19:14:08	INSERT INTO Account (Account_No, Customer_ID, Account_Type, Balance, Branch) VA
8	19:15:22	INSERT INTO Bank_Transaction (Account_No, Transaction_Type, Amount) VALUES (10
9	19:16:44	SELECT * FROM Customer LIMIT 0, 1000
10	19:17:00	SELECT * FROM Account LIMIT 0, 1000
11	19:17:16	SELECT * FROM Bank_Transaction LIMIT 0, 1000

Query9: -START A TRANSACTION

```
70 •  START TRANSACTION;
71
72
73
74
```

Output

Action Output

#	Time	Action	Message
7	19:14:08	INSERT INTO Account (Account_No, Customer_...	5 row(s) affected Reco
8	19:15:22	INSERT INTO Bank_Transaction (Account_No, ...	5 row(s) affected Reco
9	19:16:44	SELECT * FROM Customer LIMIT 0, 1000	5 row(s) returned
10	19:17:00	SELECT * FROM Account LIMIT 0, 1000	5 row(s) returned
11	19:17:16	SELECT * FROM Bank_Transaction LIMIT 0, 1000	5 row(s) returned
12	19:19:30	START TRANSACTION	0 row(s) affected

Query 10 : - DEPOSIT MONEY

```
71 •  UPDATE Account
72    SET Balance = Balance + 5000
73    WHERE Account_No = 10001;
74
```

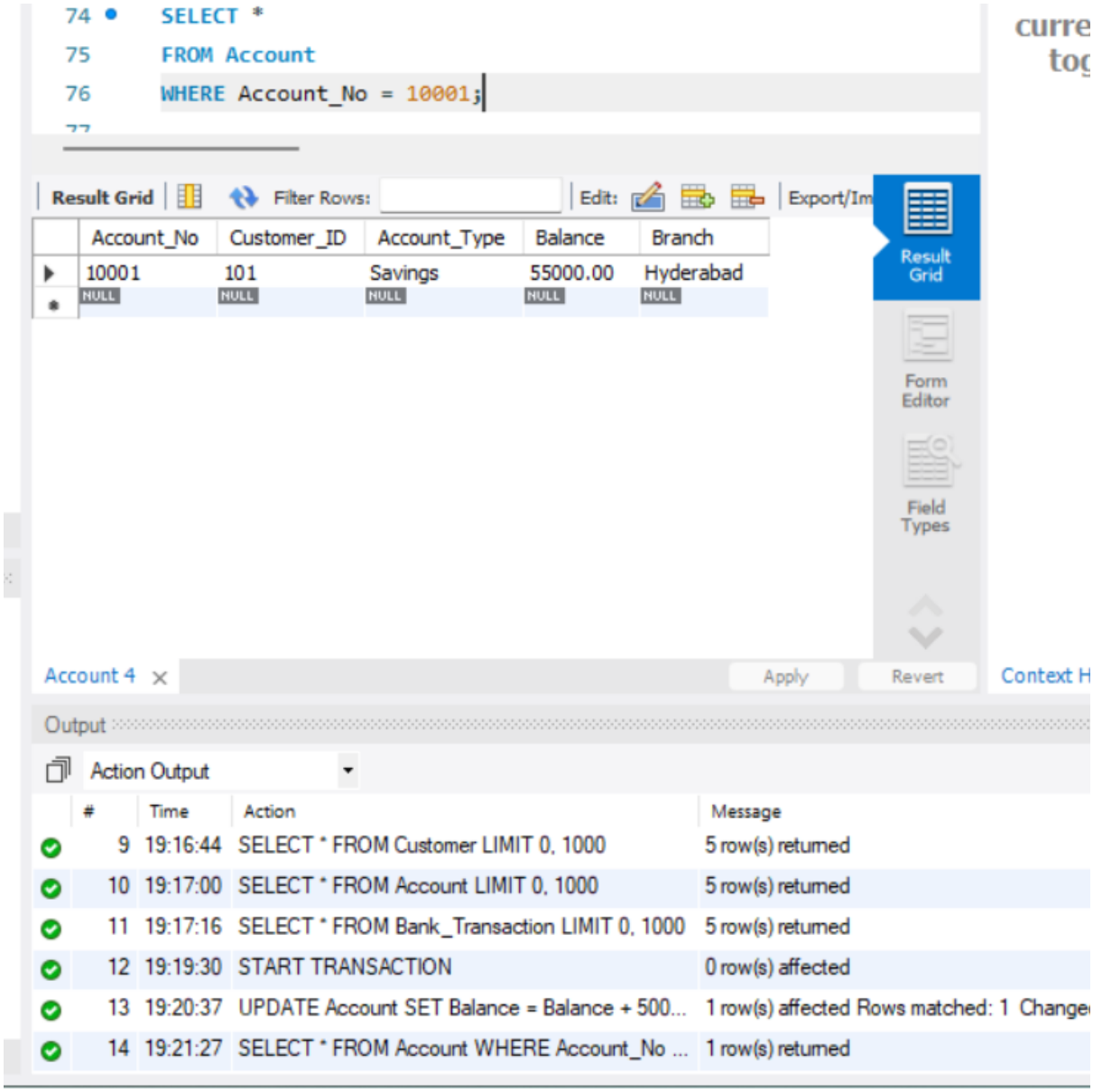
Context Help

Output

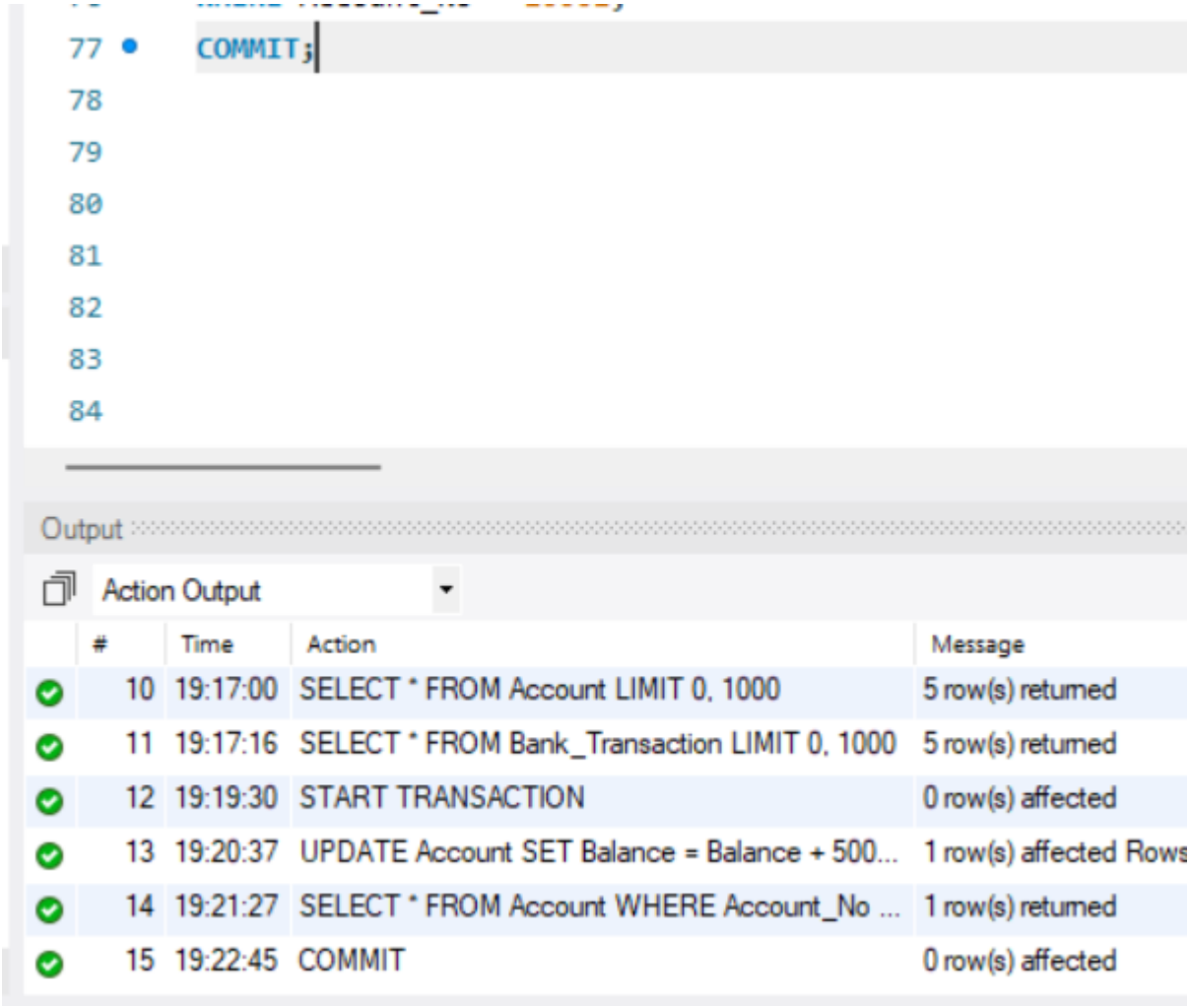
Action Output

#	Time	Action	Message
8	19:15:22	INSERT INTO Bank_Transaction (Account_No, ...	5 row(s) affected Records: 5 Duplicates: 0 Warni
9	19:16:44	SELECT * FROM Customer LIMIT 0, 1000	5 row(s) returned
10	19:17:00	SELECT * FROM Account LIMIT 0, 1000	5 row(s) returned
11	19:17:16	SELECT * FROM Bank_Transaction LIMIT 0, 1000	5 row(s) returned
12	19:19:30	START TRANSACTION	0 row(s) affected
13	19:20:37	UPDATE Account SET Balance = Balance + 500...	1 row(s) affected Rows matched: 1 Changed: 1 .

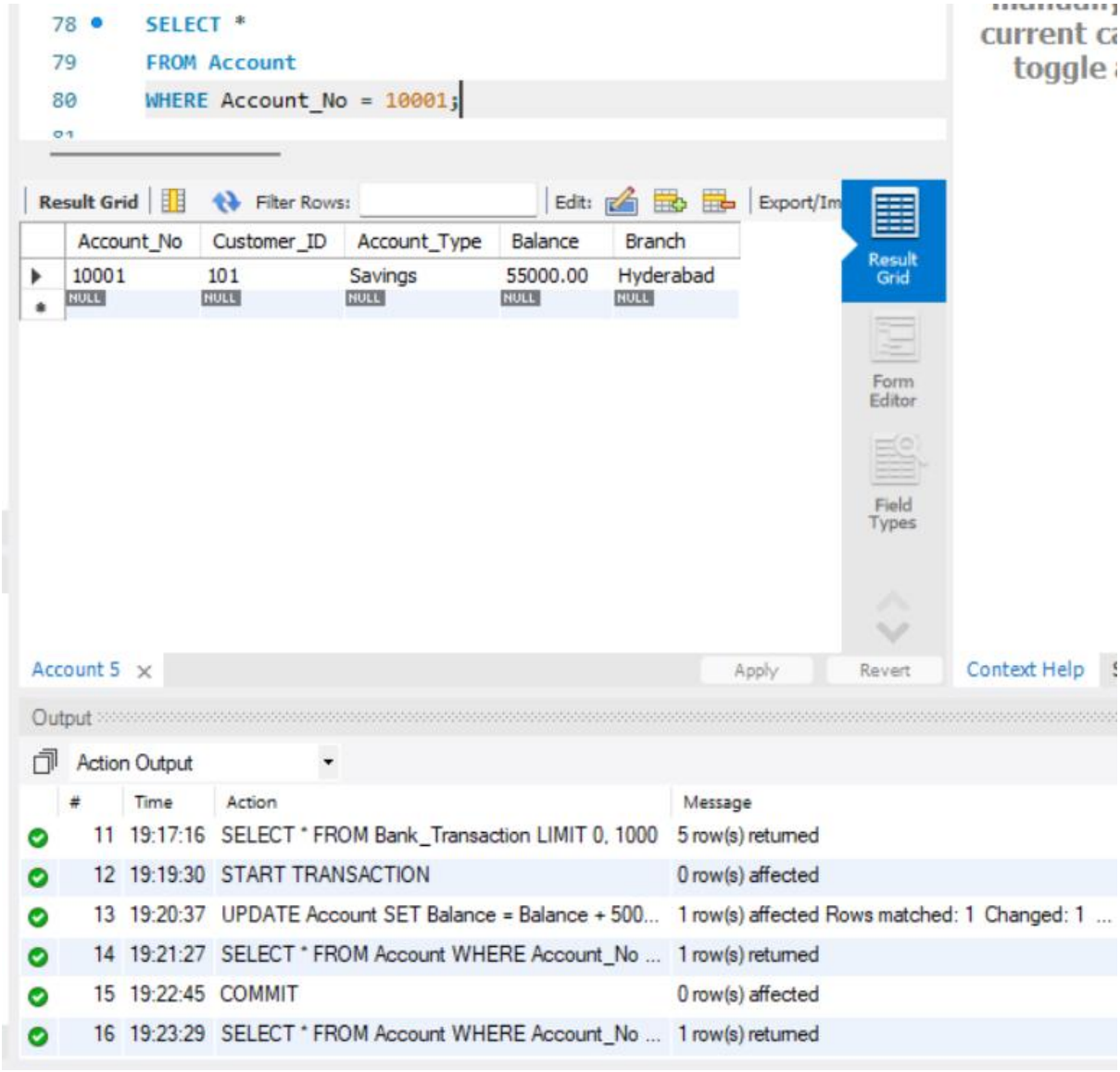
Query 11: -CHECK BALANCE



Query 12: -COMMIT TRANSACTION



Query 13: -CHECK BALANCE AFTER COMMIT



Query 14: -ROLLBACK

START TRANSACTION

```
81 •  START TRANSACTION;
82
83
84
85
86
87
88
89
90
```

Context Help | Snippets

Output

Action Output

#	Time	Action	Message	Duration / Fetch
12	19:19:30	START TRANSACTION	0 row(s) affected	0.000 sec
13	19:20:37	UPDATE Account SET Balance = Balance + 500...	1 row(s) affected Rows matched: 1 Changed: 1 ...	0.031 sec
14	19:21:27	SELECT * FROM Account WHERE Account_No ...	1 row(s) returned	0.000 sec / 0.000 sec
15	19:22:45	COMMIT	0 row(s) affected	0.031 sec
16	19:23:29	SELECT * FROM Account WHERE Account_No ...	1 row(s) returned	0.000 sec / 0.000 sec
17	19:26:05	START TRANSACTION	0 row(s) affected	0.000 sec

Query 15: -WITHDRAW MONEY

```
82 •  UPDATE Account
83    SET Balance = Balance - 10000
84    WHERE Account_No = 10001;
85
86
87
88
89
90
```

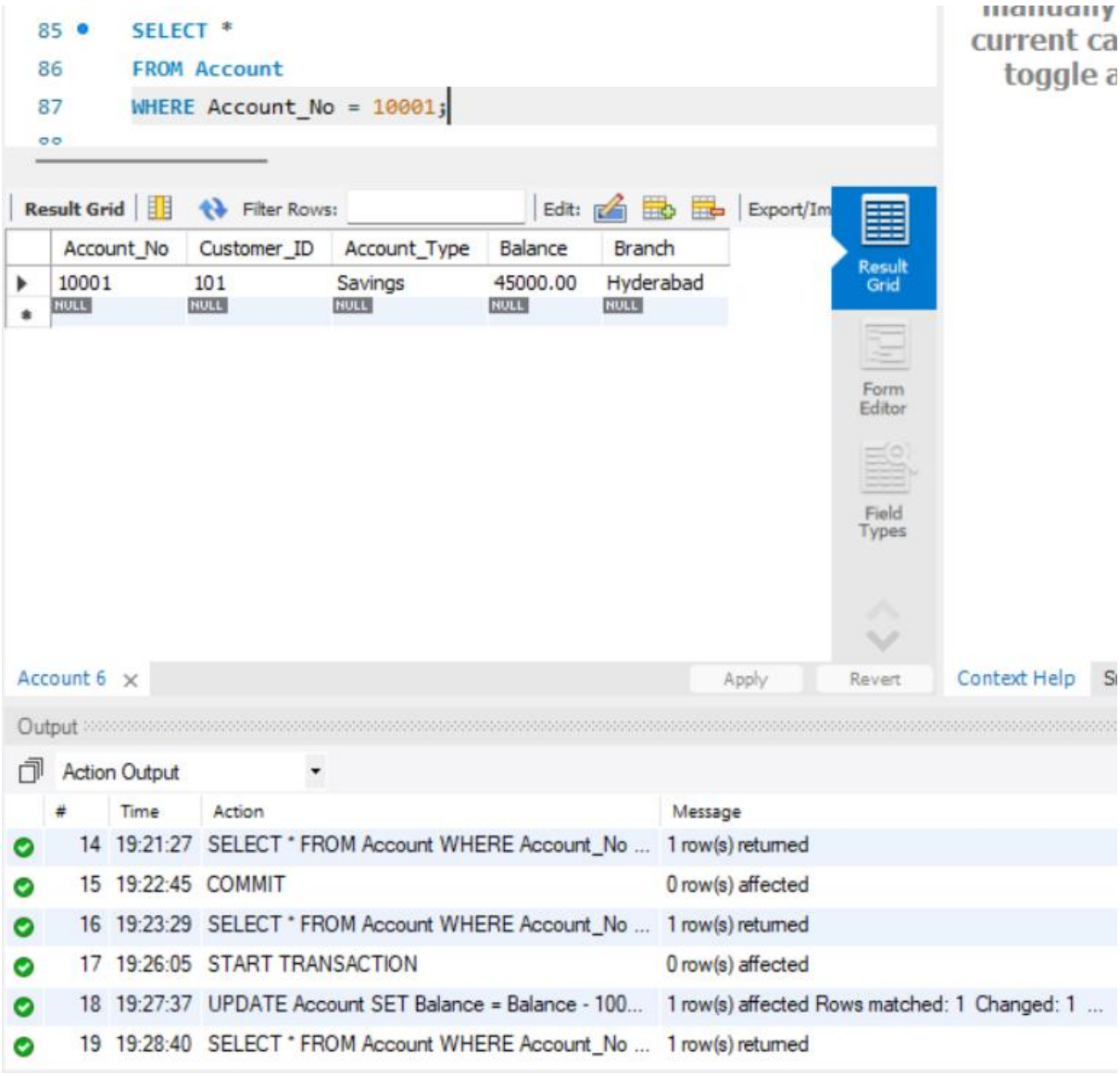
Context Help | Snippets

Output

Action Output

#	Time	Action	Message	Duration / Fetch
13	19:20:37	UPDATE Account SET Balance = Balance + 500...	1 row(s) affected Rows matched: 1 Changed: 1 ...	0.031 sec
14	19:21:27	SELECT * FROM Account WHERE Account_No ...	1 row(s) returned	0.000 sec / 0.000 sec
15	19:22:45	COMMIT	0 row(s) affected	0.031 sec
16	19:23:29	SELECT * FROM Account WHERE Account_No ...	1 row(s) returned	0.000 sec / 0.000 sec
17	19:26:05	START TRANSACTION	0 row(s) affected	0.000 sec
18	19:27:37	UPDATE Account SET Balance = Balance - 100...	1 row(s) affected Rows matched: 1 Changed: 1 ...	0.031 sec

Query 16: -CHECK BALANCE



Query 17: -ROLLBACK & CHECK BALANCE AFTER ROLLBACK

```
88 •  ROLLBACK;
89 •  SELECT *
90    FROM Account
91    WHERE Account_No = 10001;
```

Result Grid | Filter Rows: | Edit: | Export/Im

Account_No	Customer_ID	Account_Type	Balance	Branch
10001	101	Savings	55000.00	Hyderabad
NULL	NULL	NULL	NULL	NULL

Result Grid | Form Editor | Field Types

Account 7 × | Apply | Revert | Context Help

Output

Action Output

#	Time	Action	Message
16	19:23:29	SELECT * FROM Account WHERE Account_No ...	1 row(s) returned
17	19:26:05	START TRANSACTION	0 row(s) affected
18	19:27:37	UPDATE Account SET Balance = Balance - 100...	1 row(s) affected Rows matched: 1 Changed: 1 ...
19	19:28:40	SELECT * FROM Account WHERE Account_No ...	1 row(s) returned
20	19:29:31	ROLLBACK	0 row(s) affected
21	19:29:56	SELECT * FROM Account WHERE Account_No ...	1 row(s) returned

Query 18: -SAVEPOINT

```
92  START TRANSACTION;
93
94
95
96
97
98
99
100
101
102
103
104
105
106
```

Context Help | Snippets

Output

Action Output

#	Time	Action	Message	Duration / Fetch
17	19:26:05	START TRANSACTION	0 row(s) affected	0.000 sec
18	19:27:37	UPDATE Account SET Balance = Balance - 100...	1 row(s) affected Rows matched: 1 Changed: 1 ...	0.031 sec
19	19:28:40	SELECT * FROM Account WHERE Account_No ...	1 row(s) returned	0.000 sec / 0.000 sec
20	19:29:31	ROLLBACK	0 row(s) affected	0.000 sec
21	19:29:56	SELECT * FROM Account WHERE Account_No ...	1 row(s) returned	0.000 sec / 0.000 sec
22	19:31:11	START TRANSACTION	0 row(s) affected	0.000 sec

Query 19: -FIRST UPDATE

```
93  UPDATE Account
94  SET Balance = Balance + 5000
95  WHERE Account_No = 10001;
96
97
98
99
100
101
102
103
104
105
106
```

Context Help | Snippets

Output

Action Output

#	Time	Action	Message	Duration / Fetch
18	19:27:37	UPDATE Account SET Balance = Balance - 100...	1 row(s) affected Rows matched: 1 Changed: 1 ...	0.031 sec
19	19:28:40	SELECT * FROM Account WHERE Account_No ...	1 row(s) returned	0.000 sec / 0.000 sec
20	19:29:31	ROLLBACK	0 row(s) affected	0.000 sec
21	19:29:56	SELECT * FROM Account WHERE Account_No ...	1 row(s) returned	0.000 sec / 0.000 sec
22	19:31:11	START TRANSACTION	0 row(s) affected	0.000 sec
23	19:32:04	UPDATE Account SET Balance = Balance + 500...	1 row(s) affected Rows matched: 1 Changed: 1 ...	0.000 sec

Query 20: -CREATE SAVEPOINT

```
96  SAVEPOINT Deposit1;
97
98
99
100
101
102
103
104
105
106
```

Context Help | Snippets

Output

Action Output

#	Time	Action	Message	Duration / Fetch
19	19:28:40	SELECT * FROM Account WHERE Account_No ...	1 row(s) returned	0.000 sec / 0.000 sec
20	19:29:31	ROLLBACK	0 row(s) affected	0.000 sec
21	19:29:56	SELECT * FROM Account WHERE Account_No ...	1 row(s) returned	0.000 sec / 0.000 sec
22	19:31:11	START TRANSACTION	0 row(s) affected	0.000 sec
23	19:32:04	UPDATE Account SET Balance = Balance + 500...	1 row(s) affected Rows matched: 1 Changed: 1 ...	0.000 sec
24	19:33:03	SAVEPOINT Deposit1	0 row(s) affected	0.000 sec

Query 21: -SECOND UPDATE

```
UPDATE Account
SET Balance = Balance - 3000
WHERE Account_No = 10002;
```

26	19:34:14	UPDATE Account SET Balance = Balance - 300...	1 row(s) affected Rows matched: 1 Changed: 1 ...	0.000 sec

Query 22: -CREATE SECOND SAVEPOINT

```
SAVEPOINT Withdrawal1;
```

25	19:33:39	SAVEPOINT Withdrawal1	0 row(s) affected	0.000 sec

Query 23: -ROLLBACK TO SAVEPOINT

```
101 •  ROLLBACK TO SAVEPOINT Withdrawal1;
102
103
104
105
106
```

Context Help | Snippets

Output

Action Output

#	Time	Action	Message	Duration / Fetch
22	19:31:11	START TRANSACTION	0 row(s) affected	0.000 sec
23	19:32:04	UPDATE Account SET Balance = Balance + 500...	1 row(s) affected Rows matched: 1 Changed: 1 ...	0.000 sec
24	19:33:03	SAVEPOINT Deposit1	0 row(s) affected	0.000 sec
25	19:33:39	SAVEPOINT Withdrawal1	0 row(s) affected	0.000 sec
26	19:34:14	UPDATE Account SET Balance = Balance - 300...	1 row(s) affected Rows matched: 1 Changed: 1 ...	0.000 sec
27	19:36:33	ROLLBACK TO SAVEPOINT Withdrawal1	0 row(s) affected	0.000 sec

Query 24: -COMMIT

```
102 •  COMMIT;
103
104
105
106
```

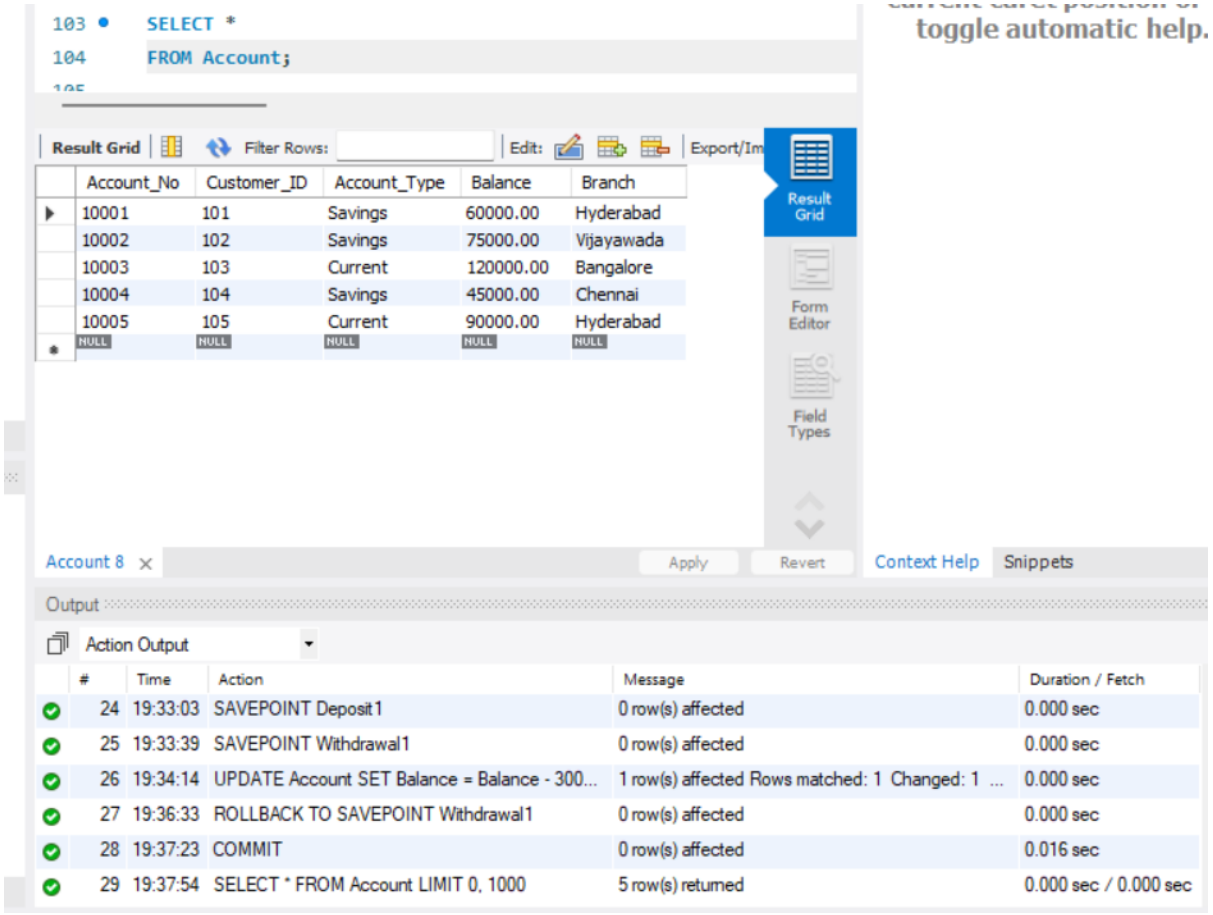
Context Help | Snippets

Output

Action Output

#	Time	Action	Message	Duration / Fetch
23	19:32:04	UPDATE Account SET Balance = Balance + 500...	1 row(s) affected Rows matched: 1 Changed: 1 ...	0.000 sec
24	19:33:03	SAVEPOINT Deposit1	0 row(s) affected	0.000 sec
25	19:33:39	SAVEPOINT Withdrawal1	0 row(s) affected	0.000 sec
26	19:34:14	UPDATE Account SET Balance = Balance - 300...	1 row(s) affected Rows matched: 1 Changed: 1 ...	0.000 sec
27	19:36:33	ROLLBACK TO SAVEPOINT Withdrawal1	0 row(s) affected	0.000 sec
28	19:37:23	COMMIT	0 row(s) affected	0.016 sec

Query 25: -CHECK ACCOUNTS


103 SELECT *
104 FROM Account;
toggle automatic help.
Result Grid
Filter Rows:
Edit:
Export/Im
Account_No Customer_ID Account_Type Balance Branch
10001 101 Savings 60000.00 Hyderabad
10002 102 Savings 75000.00 Vijayawada
10003 103 Current 120000.00 Bangalore
10004 104 Savings 45000.00 Chennai
10005 105 Current 90000.00 Hyderabad
NULL NULL NULL NULL NULL
Result Grid
Form Editor
Field Types
Account 8
Apply
Revert
Context Help
Snippets
Output
Action Output
Time Action Message Duration / Fetch
24 19:33:03 SAVEPOINT Deposit1 0 row(s) affected 0.000 sec
25 19:33:39 SAVEPOINT Withdrawal1 0 row(s) affected 0.000 sec
26 19:34:14 UPDATE Account SET Balance = Balance - 300... 1 row(s) affected Rows matched: 1 Changed: 1 ... 0.000 sec
27 19:36:33 ROLLBACK TO SAVEPOINT Withdrawal1 0 row(s) affected 0.000 sec
28 19:37:23 COMMIT 0 row(s) affected 0.016 sec
29 19:37:54 SELECT * FROM Account LIMIT 0, 1000 5 row(s) returned 0.000 sec / 0.000 sec